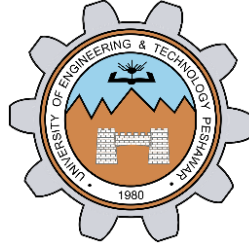


Introduction to xilinx ise and spartan 6 board

LAB # 03



Spring 2024

CSE-308L Digital System Design Lab

Submitted by: **Ali Asghar**

Registration No.: **21PWCSE2059**

Class Section: **C**

“On my honor, as student of University of Engineering and Technology, I have neither given nor received unauthorized assistance on this academic work.”

Submitted to:

Dr. Asif Ali Khan

Date:

6th March 2024

Department of Computer Systems Engineering
University of Engineering and Technology, Peshawar

Objectives:

This lab will enable students to:

- Introduction to FPGA
- Introduction to Xilinx ISE

Software used:

- Xilinx ISE

Task 1:

Implement buffer ON the Kit and attach the snapshot.

Code:

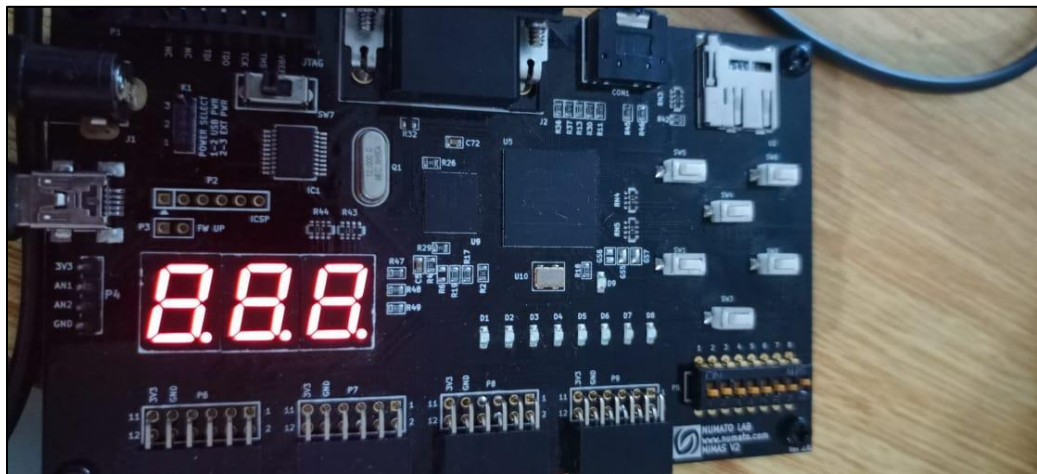
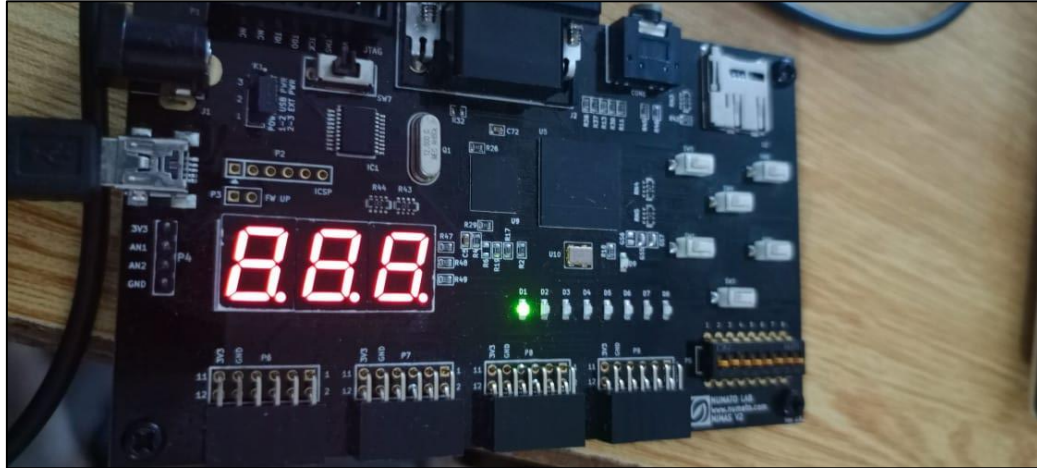
DUT:

```
21 module taskk( I, O
22     );
23     input I;
24     output O;
25     assign O = I;
26
27 endmodule
28
```

ucf file:

```
1
2 # PlanAhead Generated physical constraints
3
4 NET "A" LOC = C18;
5 NET "B" LOC = C17;
6 NET "O" LOC = P15;
7
8 # PlanAhead Generated IO constraints
9
10 NET "A" PULLUP;
11 NET "B" PULLUP;
12 NET "O" SLEW = FAST;
13
```

Output:



Task 2:

Implement AND/OR/XOR gate on the Kit and attach the snapshot.

Code:

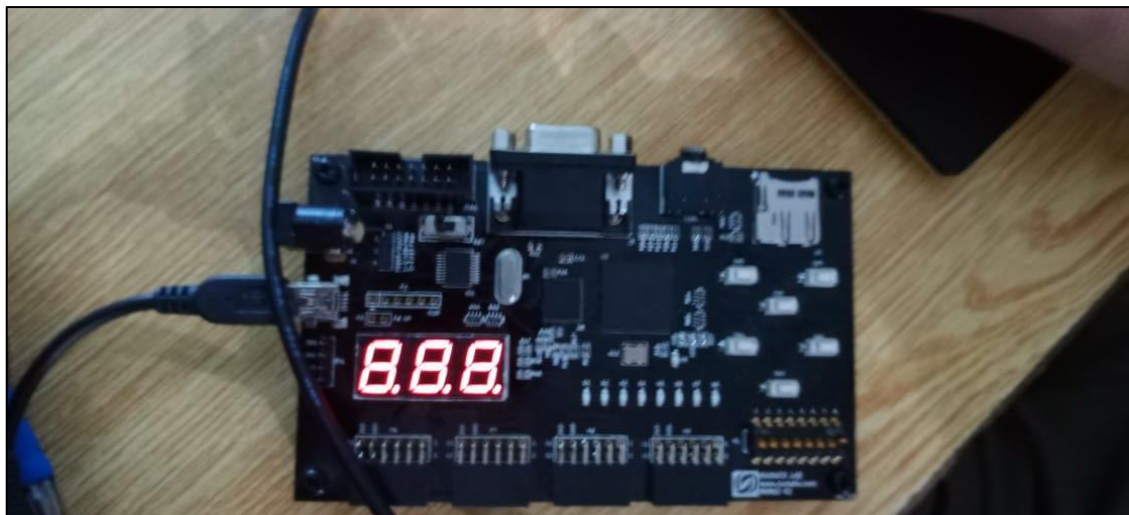
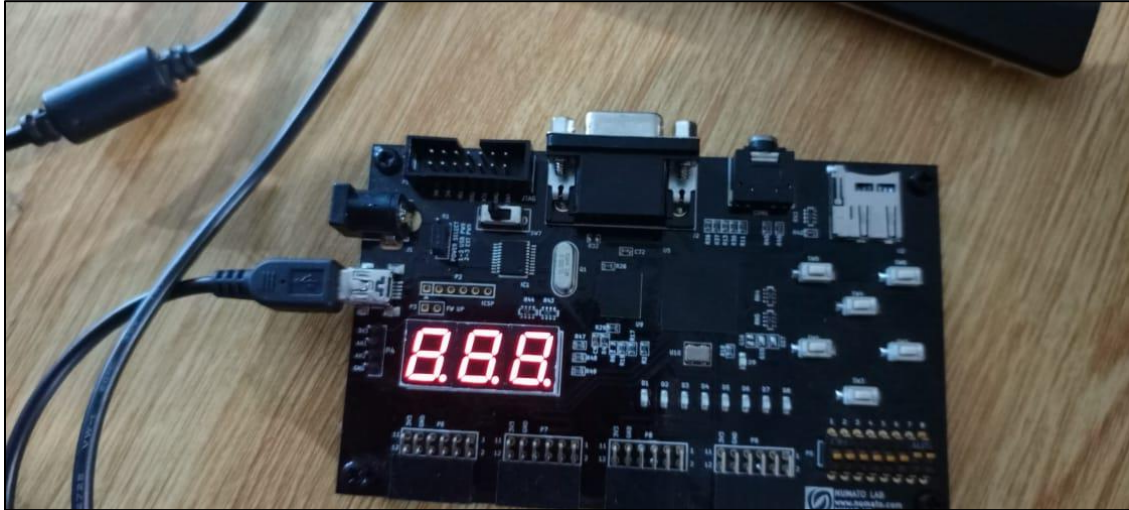
DUT:

```
21 module and_gate(A, B, O
22     );
23     input A, B;
24     output O;
25
26     assign O = A & B;
27
28 endmodule
29
```

ucf file:

```
2  # PlanAhead Generated physical constraints
3
4  NET "A" LOC = C18;
5  NET "B" LOC = C17;
6  NET "O" LOC = P15;
7
8  # PlanAhead Generated IO constraints
9
10 NET "A" PULLUP;
11 NET "B" PULLUP;
12 NET "O" SLEW = FAST;
13
```

Output:



Task 3:

Implement OR gate using NAND gate.

Code:

DUT:

```
21 module task2(A, B, O);  
22  
23     input A, B;  
24     output O;  
25     wire nA, nB;  
26  
27     nand n3(O, nA, nB);  
28     nand n1(nA, A);  
29     nand n2(nB, B);  
30  
31 endmodule
```

ucf file:

```
1  
2 # PlanAhead Generated physical constraints  
3  
4 NET "A" LOC = C17;  
5 NET "B" LOC = C18;  
6 NET "O" LOC = P15;  
7  
8 # PlanAhead Generated IO constraints  
9  
10 NET "A" PULLUP;  
11 NET "B" PULLUP;  
12 NET "O" SLEW = FAST;
```

Output:

